

# **Cinema Analytics Data Warehouse**

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# Project scope and scale

A PostgreSQL data warehouse integrating IMDb and Kaggle movie data for dimensional analysis and OLAP queries.

**3**

layers: raw → reconciled → warehouse

**45,431**

movies in the reconciled layer

**268,615**

fact rows in the warehouse

**8.04**

fact rows per included film

**7**

OLAP queries; 3 use window functions

# Source data and integration problem

Two movie datasets capture complementary aspects of the same domain; the first design decision is how to integrate them without losing source-specific meaning.

## IMDb data

- Movies are identified by the unique `tconst` key (e.g. `tt0111161` ).
- Ratings, votes, people, crew and principal roles.
- **Strong coverage** for movie identity and audience evaluation; used as the reference for ratings and roles.

## Kaggle movie metadata

- TMDb identifiers and production metadata.
- Budget, revenue, countries, languages and companies.
- **Useful for financial and production-oriented analysis**; used to complete what IMDb does not provide.

Without reconciliation, ratings and financials could not be analyzed together.

# IDs first; matching only as fallback

39,096 films carry an accepted IMDb mapping — resolved via source ID first, with exact and fuzzy title matching as controlled fallbacks; every outcome is audited.

1

**Normalize source ID**  
Canonicalize IMDb ID syntax and distinguish valid IDs from missing or invalid values.

2

**Source-ID first**  
Accept IDs found in eligible IMDb; quarantine valid source IDs that are unavailable.

3

**Controlled fallback**  
Only without a valid ID, require a unique exact title-year match, then a unique same-year fuzzy best.

4

**Audit every row**  
Record method, status and reason; never replace a valid unavailable ID heuristically.

| OUTCOME           | RECORDS |
|-------------------|---------|
| Accepted mappings | 39,096  |
| Quarantined       | 6,327   |
| Unmatched         | 10      |

Audit status governs cross-source identity, not film retention: the loader keeps 45,431 titled Kaggle films, while 39,096 carry an accepted IMDb mapping.

# Reconciled schema in third normal form

A normalized hub integrates identity, financials, ratings and production roles while preserving source relationships before warehouse denormalization.

## Core entity

The `film` table stores the unified movie identity with IMDb and TMDB references when available.

1

## Canonical relations

M:N bridges preserve roles and production data; original language is recovered and Science Fiction becomes Sci-Fi.

2

## Integrity controls

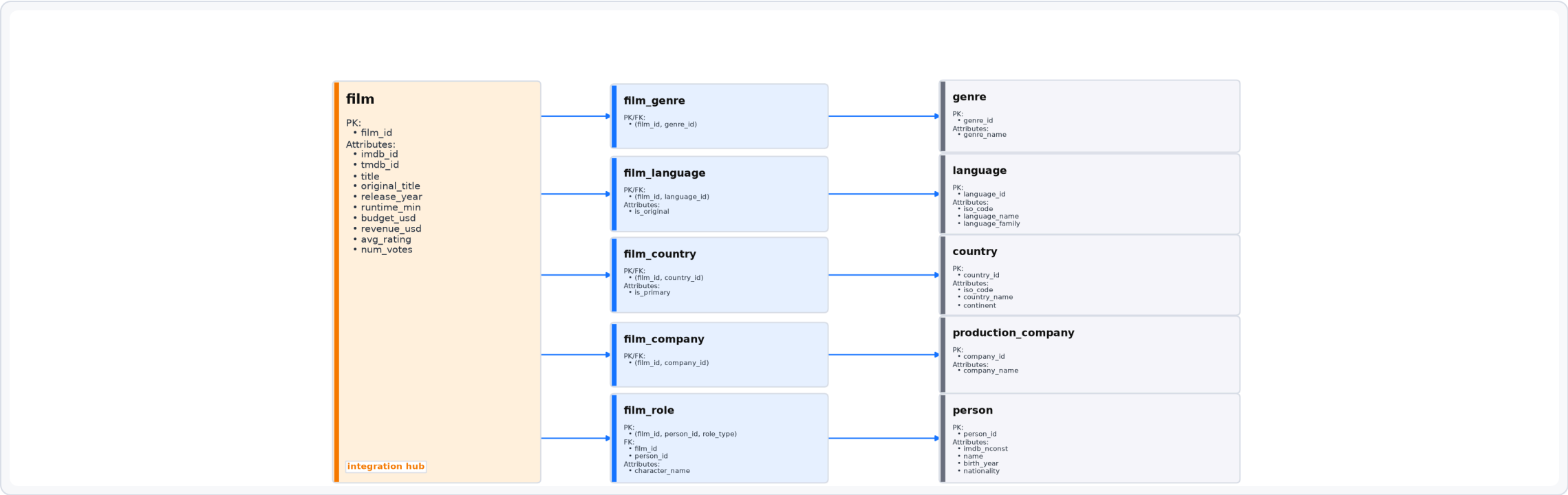
DDL constraints and ETL checks reject invalid keys, role values and source mappings.

3

| TABLE         | ROWS    | ROLE IN THE MODEL                                                 |
|---------------|---------|-------------------------------------------------------------------|
| film          | 45,431  | Cleaned Kaggle films, enriched where an IMDb mapping is accepted. |
| person        | 183,082 | Only people referenced by relevant roles of loaded films.         |
| film_role     | 474,054 | Movie-person-role relationships.                                  |
| film_genre    | 100,068 | Movie-to-canonical-genre relationships.                           |
| film_language | 59,847  | Spoken and explicitly recovered original-language links.          |

# ER diagram: 3NF reconciled layer

The separate ER view makes the normalized identity hub and its M:N relationship tables explicit.



Central **film** hub with IMDb + TMDb identifiers.

M:N bridges preserve genres, languages, countries, companies and roles.

# Fact granularity and dimensions

The DFM fixes the analytical grain before the physical star is built; that choice governs every OLAP aggregation.

## Choose analysis axes

- **Film** is degenerate; **Time** is year → decade → era; day, month and quarter are unused stubs (=1).
- **Genre**: genre → genre group.
- **ProductionContext** branches to company and country → continent.
- **Director** is flat; language → language family. Both are optional.

## Define the fact grain

- **Grain**: film × genre × production context × director × original language.
- Genre, production context and director are multiple dimensions.
- One film → multiple fact rows.

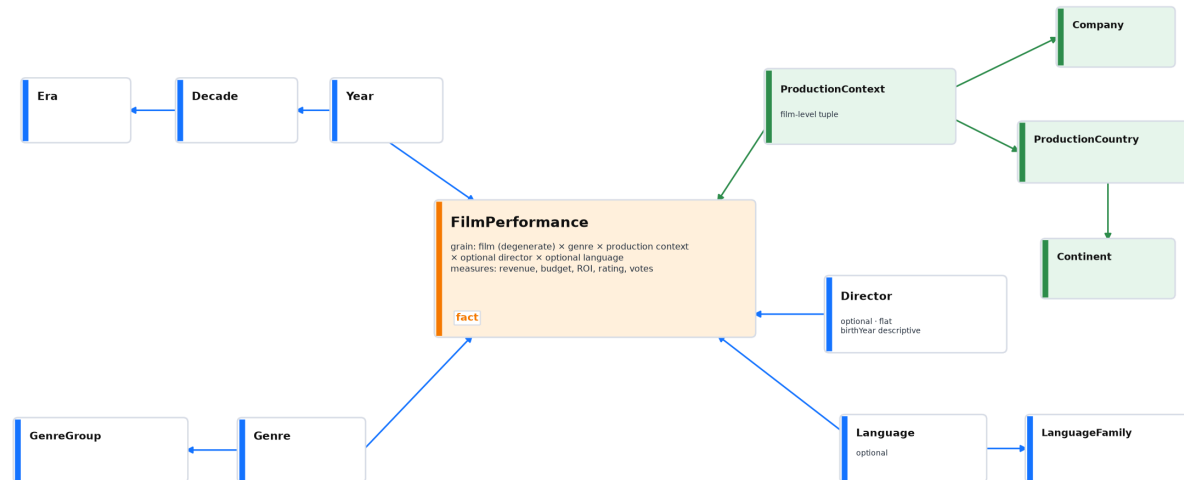
## Attach movie measures

- Revenue, budget and ROI from Kaggle.
- Average rating and vote counts from IMDb.
- Deduplicate Film before SUM / AVG; never SUM ROI or rating.

Genre, ProductionContext and Director create fan-out. Director and original language are optional; original language is single and does not multiply rows.

# DFM diagram: fact grain and hierarchies

The conceptual model separates the fact, aggregation semantics and analysis coordinates before physical implementation.



**ProductionContext** branches independently to Company and ProductionCountry → Continent; it never asserts company domicile.

# Star schema implementation

Denormalized dimensions keep roll-up, drill-down, slice and dice paths readable around `fact_film_performance`.

## Star schema

Five denormalized dimensions around the fact.



Short, readable OLAP join paths.

1

## ETL eligibility

Year 1900–2030, at least one genre and one resolvable production company.



33,418 / 45,431 films pass the filter (73.6%).



73.6% fact coverage; 26.4% excluded only by ETL policy.

2

## Optional axes

Production context may have no country; Director and Language FKs are nullable.

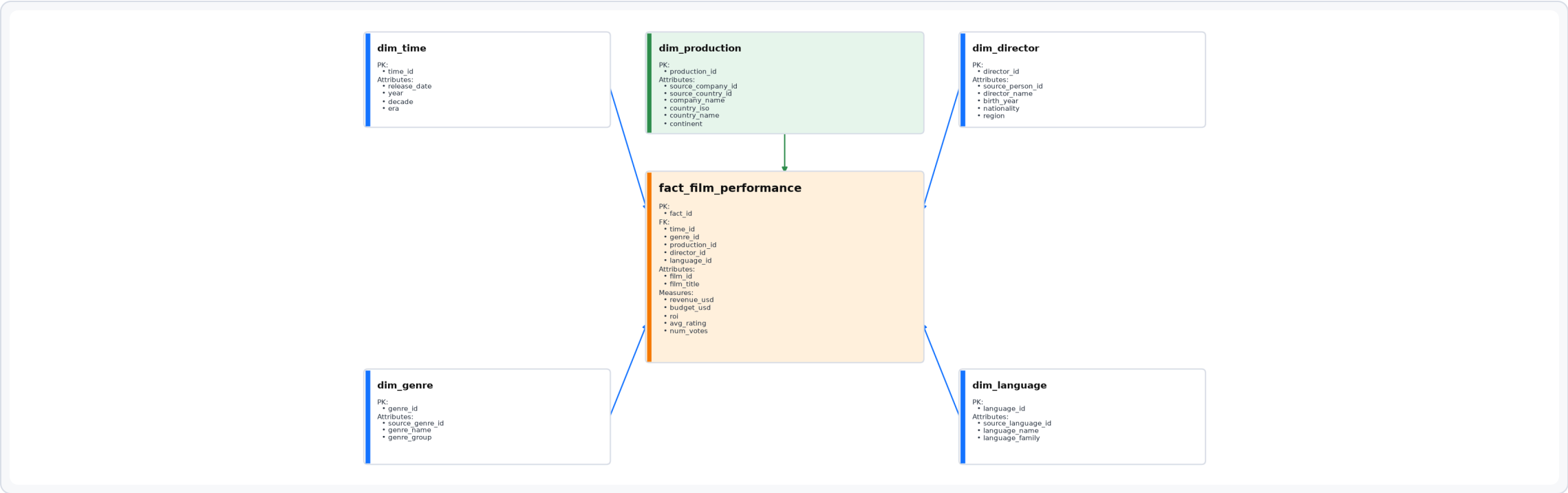


Missing optional data never excludes an eligible film.

3

# Star schema: OLAP-ready implementation

The physical warehouse places the film-performance fact at the center of five analytical dimensions.



268,615 fact rows, 33,418 distinct films.

5 dimension tables plus degenerate Film. DDL constraints + ETL preflight/checks: local/row-level integrity.

48,609 / 49,150 production rows have a continent; manual delivery-snapshot checks: cross-row measure consistency + fact coverage.

# Deduplication prevents inflated totals

## Why the bias appears

- Genres, company-country contexts and directors are M:N — they fan-out fact rows.
- Original language is single and optional, so it does **not** create fan-out.
- Revenue and budget are **movie-level** measures.
- Summing across fan-out rows **inflates** those values.



## How analytical validity is preserved

- Use a CTE to deduplicate at the target dimensional level.
- Aggregate only **after deduplication**.
- Apply consistently: results must reflect real data, not modeling artifacts.



| EXAMPLE                            | WITHOUT DEDUPLICATION                                                        | WITH DEDUPLICATION                                                           |
|------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| One film: Furious 7                | Its revenue is repeated in every genre × company-country × director row.     | Collapse the repeated rows into <b>one contribution</b> at the target grain. |
| Genre–decade result: Action, 2010s | A direct SUM adds those repeated revenues, <b>inflating</b> the group total. | Keep one row per film, decade and genre group, then aggregate.               |

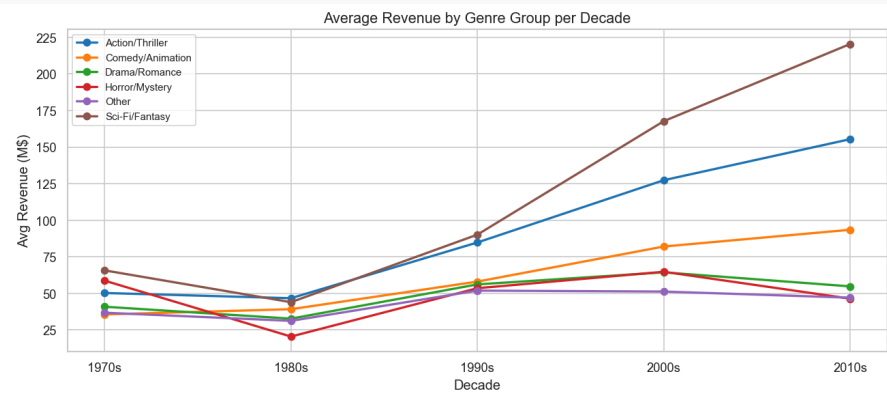
# Seven queries cover the required OLAP patterns

Four classical OLAP queries and three window-function analyses.

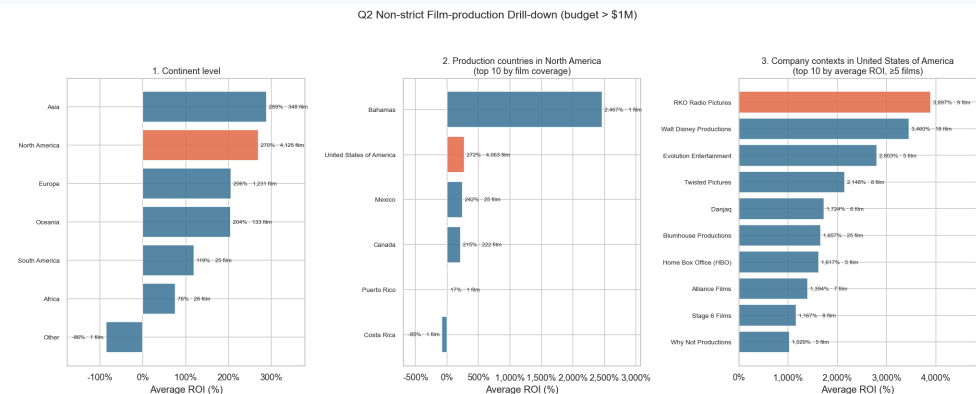
| QUERY                     | OPERATION           | ANALYTICAL QUESTION                   |
|---------------------------|---------------------|---------------------------------------|
| Classical OLAP operations |                     |                                       |
| Q1                        | Roll-up             | Revenue by genre group and decade     |
| Q2                        | Drill-down + slice  | ROI from continent to company context |
| Q3                        | Dice                | Ratings by era and language family    |
| Q4                        | Slice + CTE         | ROI by studio size                    |
| Window-function queries   |                     |                                       |
| Q5                        | NTILE + percentiles | ROI across budget quartiles           |
| Q6                        | SUM OVER            | Cumulative revenue by genre           |
| Q7                        | RANK                | Top-rated films by decade             |

# Visual outputs for Q1–Q4

The first four plots connect roll-up, drill-down, dice and slice operations to the current canonical CSV exports.



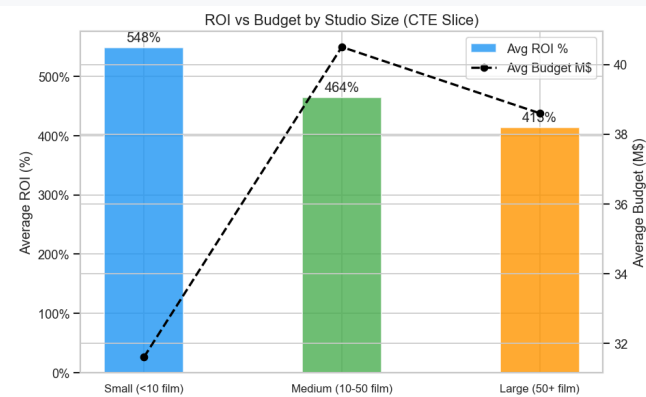
Q1 • Revenue by genre group and decade



Q2 • Production drill-down



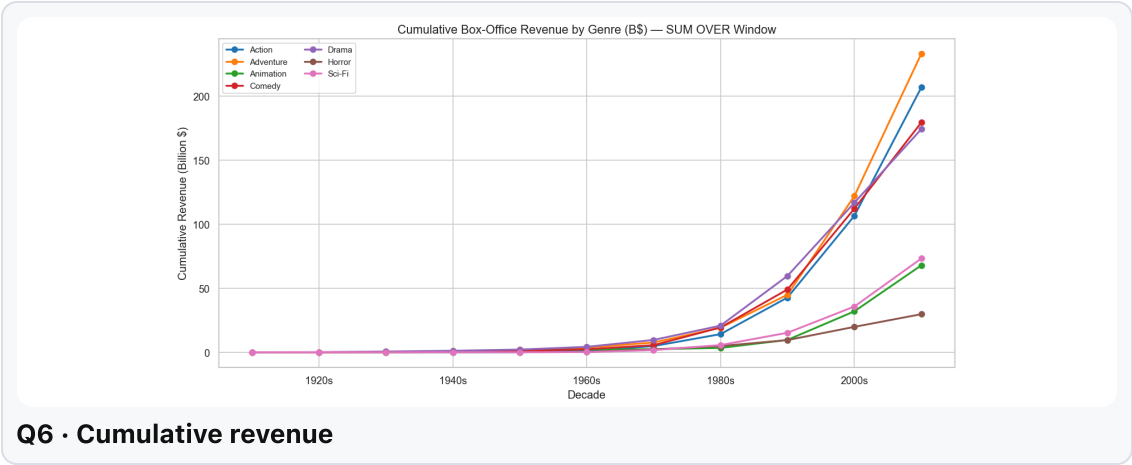
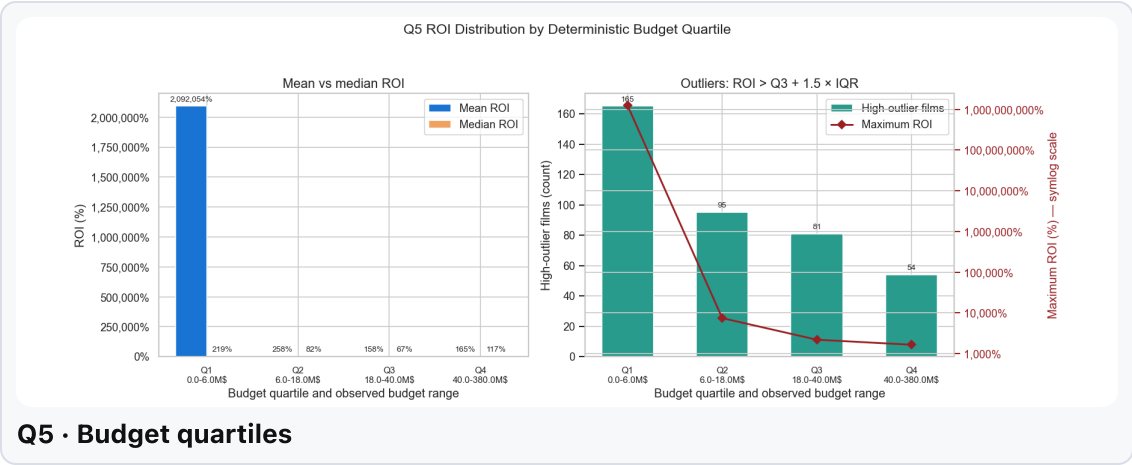
Q3 • Rating by era and language family



Q4 • ROI by studio size

# Window-function outputs for Q5–Q7

The remaining plots show NTILE budget bands, independent cumulative SUM OVER series and RANK by decade.



Top-5 Films by IMDb Rating per Decade (RANK() OVER PARTITION BY decade)

| Rank | 1950s                                                  | 1960s                                                     | 1970s                                       | 1980s                                                           | 1990s                                     | 2000s                                                              | 2010s                                                            | 2020s                                       |
|------|--------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------|
| #1   | 12 Angry Men (9.0 ★, 989k v)                           | The Good, the Bad and the Ugly (8.8 ★, 893k v)            | The Godfather (9.2 ★, 2234k v)              | The Marathon Family (8.8 ★, 18k v)                              | The Shawshank Redemption (9.3 ★, 3202k v) | The Dark Knight (9.1 ★, 3183k v)                                   | O.J.: Made in America (8.9 ★, 24k v)                             | Avatar: The Way of Water (7.5 ★, 634k v)    |
| #2   | Seven Samurai (8.6 ★, 403k v)                          | The Human Condition III: A Soldier's Prayer (8.6 ★, 9k v) | The Godfather Part II (9.0 ★, 1499k v)      | The Moromete Family (8.8 ★, 5k v)                               | Schindler's List (9.0 ★, 1590k v)         | The Lord of the Rings: The Return of the King (9.0 ★, 2172k v)     | Inception (8.8 ★, 2829k v)                                       | Bad Boys for Life (6.5 ★, 198k v)           |
| #3   | The Counterfeit Coin (8.6 ★, 3k v)                     | World Gone Mad (8.8 ★, 3k v)                              | Tosun Pasha (8.9 ★, 26k v)                  | The Kingdom of Diamonds (8.8 ★, 3k v)                           | Sandesham (9.0 ★, 6k v)                   | The Lord of the Rings: The Fellowship of the Ring (8.9 ★, 2214k v) | The Phantom of the Opera at the Royal Albert Hall (8.8 ★, 11k v) | The King's Daughter (5.2 ★, 7k v)           |
| #4   | The Human Condition I: No Greater Love (8.5 ★, 11k v)  | Harakiri (8.6 ★, 92k v)                                   | The Message (8.9 ★, 12k v)                  | Star Wars: Episode V - The Empire Strikes Back (8.7 ★, 1510k v) | Fight Club (8.8 ★, 2623k v)               | The Lord of the Rings: The Two Towers (8.8 ★, 1964k v)             | Natsamrat (8.8 ★, 8k v)                                          | The Prey: Legend of Karnoctus (3.6 ★, 1k v) |
| #5   | The Human Condition II: Road to Eternity (8.5 ★, 9k v) | Psycho (8.5 ★, 788k v)                                    | The Chaos Class Is Waking Up (8.7 ★, 22k v) | Stop Making Sense (8.7 ★, 24k v)                                | Forrest Gump (8.8 ★, 2506k v)             | Before Flying Back to Earth (8.7 ★, 1k v)                          | Interstellar (8.7 ★, 2550k v)                                    | —                                           |

Q7 · Top films by decade

# What the model supports — and where it stops

Stable invariants protect the model; documented source limits define the boundary of interpretation.

## What is demonstrated

- Reconciled and warehouse layers run in PostgreSQL; raw sources stay files, and OLAP queries run over the warehouse.
- Stable integrity checks block invalid states.
- Canonical exports connect every query to its visual output.

## Documented limitations

- Director nationality is unavailable in IMDb.
- Production geography is film context, not company domicile.
- Budget coverage limits the financial subset.

# Reliable analysis starts before the query

## Reconcile identity

Join complementary sources through auditable IDs.

## Fix the grain

Define what one fact row represents before aggregation.

## Deduplicate first

Aggregate movie-level measures only at the target analytical level.

## Summary

The project demonstrates a complete analytical pipeline — two heterogeneous sources, a reconciled integration layer, a dimensional star schema, and seven OLAP queries over the current fact snapshot.

thank you